

Technology Survey For The Lost & Found Item Tracker

Group 8

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1. Frontend Selection & Justification

1.1 React JavaScript Library

React, also known as React.js or ReactJS, is an open-source front-end software library, with its primary programming language being JavaScript. React allows developers to build applications using components, which are small, reusable pieces of the user interface, of which can be buttons, forms, navigation bars, search bars, drop-downs, and any other constituents. Each component manages its own state and is able to combine to form complex interfaces, providing real-time updates to the application as data is modified during development.

1.2 Vue.js

Vue is a frontend framework that utilizes JavaScript for its functionality. Vue is marketed as an approachable, performant, and versatile framework that is scalable for web user interfaces. It was created by a former Google employee who wanted to improve upon the Angular framework Google developed taking the positives from Angular and building something lightweight.

Vue builds on top of standard HTML, CSS, and JavaScript with an intuitive API and documentation. It utilizes a reactive rendering system to be able to visually see how changes to the code impact the elements. Vue is also designed to scale with additional libraries and frameworks to better customize the system for the developer's needs.

1.3 Angular

Angular is an open-source JavaScript-based frontend platform that is supported by Google. It follows a Model-View-Controller architecture for designing a web application that allows each part to be developed in parallel and separately tested, allowing the code to be easier to understand and maintain.

Angular largely uses TypeScript, a syntactical superset of Javascript developed by Microsoft for large applications, instead of JavaScript. Angular is also designed with mobile browsers in mind to provide support for mobile web and native mobile applications.

1.4 Comparison Table

Software	Pros	Cons
React JavaScript Library	React's component-driven design encourages modular, reusable UI elements. This, in turn, further enhances code organization, consistency, reusability, scalability, and maintainability across larger projects. The technology has highly efficient performance with the Virtual Document Object Model as React's DOM minimizes costly direct interactions with the real DOM, of which only the necessary parts of the UI are re-rendered, improving application speed and responsiveness. React also has a strong ecosystem and community support, proven by its extensive documentation and with thousands of third-party libraries and developer tools. With an active community, it ensures quick issue resolution, frequent updates, and abundant learning resources.	While React itself is simple, setting up a full-featured app (with routing, state management, build tools, etc.) can involve complex configuration, which at times leads to developers needing to combine multiple libraries to achieve framework-level functionality. The React ecosystem evolves quickly, with certain third-party tools becoming obsolete or incompatible over time, requiring developers to stay updated with frequent library updates, new patterns, and caution for potential outdated documentation. JSX can be confusing or unappealing for developers who prefer separating HTML and JavaScript logic, adding a small learning overhead compared to template-based frameworks like Vue or Angular.
Vue	Vue is flexible and allows developers to use it as a library for specific features or as a full-fledged framework for complex applications. Vue allows for defining components in a single file, which includes the template, script, and styles, making it easier to manage components. Vue is known for its simplicity and approachability, making it easy for beginners to understand and use.	Vue is a relatively young technology compared to React and Angular, as such there are less experienced developers for the technology. Vue has a risk of over flexibility which could give a development team too many options that might lead to different programming approaches occurring within one team.
Angular	Angular is a full-feature framework with a comprehensive set of tools and libraries for	Angular is verbose and complex due to the way that components are managed, possibly requiring up to

Angular	building complex web applications. Angular follows the Model-View-Controller (MVC) architecture, which helps in organizing and managing code. Angular has a built-in dependency injection system that helps in managing dependencies and promoting modular code.	five files for a single component with the necessary dependencies. Angular has a difficult learning curve compared to React or Vue, as one would have to learn TypeScript in addition to the modules, dependency injection, and components of Angular.
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1.5 Final Selection

The team has decided to utilize React for the frontend framework, as members involved in its development are most familiar due to prior experience with the software. With React's reusable structure, encapsulation, vast ecosystem of high-quality third-party libraries, and extensive documentation, this approach enhances code and architecture comprehension, consistency, and long-term maintainability throughout the development of the Lost and Found Item Tracker application.

2. Database Selection & Justification

2.1 PostgreSQL

PostgreSQL is a powerful, open-source relational database management system (RDBMS) known for its strong adherence to SQL standards, full ACID compliance, and extensive feature set. It supports advanced data types and robust indexing options, making it ideal for complex applications that require flexibility and reliability. PostgreSQL integrates exceptionally well with Django, offering built-in support for features not found in other database management systems.

From a development perspective, PostgreSQL offers powerful tools for migrations and backup management, both of which are useful in a collaborative project setting. Its active community and detailed documentation make it a dependable choice for long-term maintenance. The primary drawbacks are slightly higher setup complexity and resource consumption compared to lighter databases such as MySQL or SQLite. However, its reliability, standards compliance, and compatibility with Django make it the preferred option for scalable web applications such as the Lost & Found Item Tracker.

2.2 MySQL

MySQL is one of the most widely used open-source databases and is well-known for its speed, simplicity, and ease of deployment. It is a popular choice for web applications due to its straightforward configuration, efficient query performance for read-heavy workloads, and strong support across major hosting services. Django supports MySQL natively, and it performs well for typical applications with moderate complexity.

MySQL has several limitations, however, that can impact long-term scalability and data integrity. MySQL's query planner and indexing system are efficient for common queries but less flexible for complex joins or analytical workloads. Additionally, some advanced Django features, like partial indexes and native JSON operations, are limited in MySQL compared to PostgreSQL.

Despite these drawbacks, MySQL remains a reasonable choice for lightweight or mid-scale web applications where ease of setup and fast read performance are higher priorities than strict standard compliance or advanced data modeling.

2.3 Oracle

Oracle Database is an enterprise-grade RDBMS designed for large-scale applications that demand high reliability, extensive optimization capabilities, and advanced security features. It offers powerful transaction handling, fine-grained access control, and performance tuning tools that make it suitable for mission-critical systems. Oracle provides strong support for SQL standards, as well as integration options for large enterprise infrastructures.

However, Oracle's commercial licensing model and high administrative complexity make it impractical for most academic or small-team projects. Setting up and maintaining an Oracle instance requires more resources and expertise compared to open-source alternatives. Although Django includes official support for Oracle, it is rarely used in educational or prototype environments due to the high cost and heavy configuration overhead.

For a project such as the Lost & Found Item Tracker, Oracle would be excessive in both scale and cost, offering little practical advantage over open-source systems like PostgreSQL that already meet the project's needs for reliability, consistency, and scalability.

2.4 Comparison Table

Database	Pros	Cons
PostgreSQL	PostgreSQL offers high support in Django, as many Django features are only fully supported in PostgreSQL. PostgreSQL also features strong data integrity and reliability. PostgreSQL supports complex queries and powerful indexing PostgreSQL is highly scalable, offering excellent performance for large datasets and multiple-user workloads.	PostgreSQL is more complex than other considered options, which can make the barrier for entry higher for group members with less experience in database management. PostgreSQL has higher memory usage, which might not be suitable for small-scale database designs.
MySQL	MySQL has a long history of use in web applications, and is stable and reliable. MySQL has very good performance for simple queries, making it ideal for smaller scale projects. MySQL is easy to set up and is versatile, being supported by many different hosting providers.	MySQL has limited feature support for Django, which can limit the operations done on the database. MySQL has weaker data integrity than other considered options, and might not be as trustworthy for larger scale projects.
Oracle	Oracle is highly robust with excellent scalability, promising strong performance. Oracle has excellent reliability and security features, ensuring strong data integrity and security. Oracle comes with official Django backend support, allowing it to integrate well into web applications.	Oracle features a complex setup, requiring extensive configuration and licensing. Oracle can be costly, as it is a commercial software that comes with licensing fees. Oracle can be overkill for smaller projects such as academic or small-team projects.

2.5 Final Selection

After evaluating PostgreSQL, MySQL, and Oracle, the team selected PostgreSQL as the database technology for the Lost & Found Item Tracker project. PostgreSQL provides the best balance of reliability, scalability, and compatibility among the options considered. Its advanced feature set, such as native support for JSON data and robust indexing capabilities, aligns well with the project's need for accurate item tracking, search

functionality, and multi-user interaction. Additionally, PostgreSQL's open-source nature and active community ensure both accessibility and long-term maintainability. While MySQL offers simplicity and performance for smaller-scale applications, it lacks some of the advanced capabilities needed for future feature expansion. Oracle, though powerful, is cost-prohibitive and unnecessarily complex for this scope. Overall, PostgreSQL offers the most practical and technically sound foundation for building a reliable, scalable, and maintainable web application.

3. Backend Selection & Justification

3.1 Django

Django is an open-source, high-level Python Web Framework widely used for building secure and scalable web applications. It follows the Model-View-Template (MTV) architecture and includes built-in tools for tasks like authentication, database management, and form handling, allowing developers to create robust applications quickly. Django's strong emphasis on security helps protect against common web vulnerabilities such as SQL injection and cross-site scripting.

Django's modular structure and object-relational mapping (ORM) system enables efficient data management and seamless integration with databases like PostgreSQL. Using our Lost & Found Item Tracker project, for example, Django would simplify the implementation of features we will add to ensure data consistency and reliability. Its well organized codebase and extensive documentation makes the application simple to maintain, test, and debug.

3.2 Node.js

Node.js is an open-source JavaScript runtime environment well known for its use in the development of servers and web applications. It is easily scalable due to its non-blocking, event-driven architecture, which makes it useful for handling simultaneous connections with real-time data processing. Node.js also has existing React integration along with PostgreSQL integration libraries that would allow for the quick and easy implementation of secure data management and features like user authentication.

From a development perspective, the non-blocking, event-driven architecture of Node.js would ensure efficient performance while carrying out data-heavy operations in our Lost & Found Item Tracker, such as item claim updates and item searching. Node.js would also be very convenient for the application's maintainability and debugging, since it has the benefit of using JavaScript for both the frontend and backend development.

3.3 Comparison Table

Framework	Pros	Cons
Django Web Framework	Its speed and scalability allows it to build and scale up projects incredibly quickly which is why it is popular in tech startups. Django has the best sets of documentation among open source frameworks out there. With well organized, beginner friendly, and practical examples. Django has a built-in security that prevents it from common threats and attacks such as XXF, CSRF, and SQL injections, as well as HTTPS. Django has great versatility so it is compatible with a wide range of applications from content management systems or an entire social network. It can even carry out machine learning or data analysis.	Django's high-level framework is made for more complex projects and applications. Simpler ones are often better off built with simpler frameworks such as Flask. Large scale projects with many database interactions may lead to slower performance if not optimized properly. Before building the code developers must build the whole structure out due to its complexity. Django does not have any conventions, a rulebook of sorts, which can lead to slower developments or less standardized.
Node.js	Node.js's non-blocking event-driven architecture is beneficial for scalability and the handling of simultaneous connections with real-time data processing. Node.js allows for full-stack JavaScript development, handling both the front and backend with JavaScript. Node.js's default package manager provides over 2 million module libraries, which would allow for fast and easy integration of features like authentication.	Node.js is not optimal for heavy CPU tasks since it operates on a single-thread event loop, which could impact performance if improperly handled. The asynchronous programming model that Node.js provides could result in deeply nested callbacks if unfamiliar with the framework, impacting the code's readability and maintainability. Node.js's rapid ecosystem can cause its module libraries to become outdated or

		deprecated which could result in surprise problems with previously functioning code.
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3.4 Final Selection

After evaluating Django and Node.js, the team has decided to work with the Django Web Framework for the Lost & Found Item Tracker. Much of the team has prior experience with Django and is familiar with its built-in tools, which will streamline the implementation of core requirements like user verification and form handling. Django is also compatible with PostgreSQL and can be easily integrated to work with React via the Django REST framework, allowing for the quick and reliable development, scalability, and maintainability of the Lost & Found Item Tracker application.

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